预备知识

泰勒展开

$$e^x, a^x, \ln(1+x) \qquad \frac{1}{1+x}, \frac{1}{1-x} \qquad \sin x, \cos x, \arctan x \qquad (1+x)^\alpha \qquad \tan x, \arcsin x$$

$$x_0 = 0, x \to 0\text{时}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \circ(x^n)$$
$$\mathrm{a}^x = e^{x\ln a} = 1 + x\ln a + \frac{(x\ln a)^2}{2!} + \cdots + \frac{(x\ln a)^n}{n!} + \circ(x^n)$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + \frac{(-1)^{n-1}x^n}{n} + \circ(x^n)$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-1)^n x^n + \circ(x^n)$$
$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \circ(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + \frac{(-1)^n}{(2n+1)!}x^{2n+1} + \circ(x^{2n+1})$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + \circ(x^{2n})$$
$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + \frac{(-1)^n}{2n+1}x^{2n+1} + \circ(x^{2n+1})$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \circ(x^n) \qquad (\alpha\text{为任意常数，此式可无限展开})$$

记前两项：

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots + \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!}x^{2n-1} + \circ(x^{2n-1})$$
$$\arcsin x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots \frac{(2n)!}{4^n(n!)^2(2n+1)}x^{2n+1} + \circ(x^{2n+1})$$

数列重要公式

等差、等比求和

$$1^2 + 2^2 + 3^3 + \cdots + n^2$$

$$S_n = \frac{n(a_1 + a_n)}{2}$$

$$S_n = \frac{a_1(1-q^n)}{1-q}$$

$$1^2 + 2^2 + 3^3 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

三角函数常用公式

$$(\sin^2 x, \cos^2 x), (\tan^2 x, \sec^2 x), (\cot^2 x, \csc^2 x)$$

$$\sin 2\alpha, \cos 2\alpha$$

$$\sin(\alpha \pm \beta), \cos(\alpha \pm \beta), \tan(\alpha \pm \beta)$$

$\sin^2 x + \cos^2 x = 1$	$1 + \tan^2 x = \sec^2 x$	$1 + \cot^2 x = \csc^2 x$
$\sin 2\alpha = 2\sin\alpha\cos\alpha$	$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$	
$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$	$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$	$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$

积化和差、和差化积

$$\sin\alpha\cos\beta,\cos\alpha\sin\beta,\cos\alpha\cos\beta,\sin\alpha\sin\beta$$

$$\sin\theta+\sin\gamma,\sin\theta-\sin\gamma,\cos\theta+\cos\gamma,\cos\theta-\cos\gamma$$

积化和差：

$$\sin\alpha\cos\beta=\tfrac{1}{2}[\sin(\alpha+\beta)+\sin(\alpha-\beta)]$$

$$\cos\ \alpha\sin\beta=\tfrac{1}{2}[\sin(\alpha+\beta)-\sin(\alpha-\beta)]$$

$$\cos\ \alpha\cos\beta=\tfrac{1}{2}[\cos(\alpha+\beta)+\cos(\alpha-\beta)]$$

$$\sin\ \alpha\sin\beta=\tfrac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)]$$

和差化积（由积化和差公式反得）：

$$\theta=(\alpha+\beta),\gamma=(\alpha-\beta)$$
$$\begin{cases}\text{①拆项整理}\\ \text{②反解变量}\begin{cases}\alpha=\frac{\theta+\gamma}{2}\\ \beta=\frac{\theta-\gamma}{2}\end{cases}\end{cases}$$

因式分解公式

$$a^3-b^3,a^3+b^3$$

$$a^3-b^3=(a-b)(a^2+ab+b^2)$$

$$a^3+b^3=(a+b)(a^2-ab+b^2)$$

常用不等式（等号成立条件）

不等式灵活使用，形式多样本质不变，勿固化

$$||a|-|b||\leqslant|a\pm b|\leqslant|a|+|b|$$

$$\frac{2}{\frac{1}{a}+\frac{1}{b}}\leqslant\sqrt{ab}\leqslant\frac{a+b}{2}\leqslant\sqrt{\frac{a^2+b^2}{2}}(a,b>0,a=b\text{时等号成立;}\quad\text{高维同样成立})$$
$$\color{blue}{\text{注：灵活取平方}}$$

$$\begin{array}{l}\sin x\leqslant x\leqslant\tan x\\ \arctan x\leqslant x\leqslant\arcsin x\\ \ln(x+1)\leqslant x\leqslant e^x-1\end{array}$$

$$\frac{1}{1+x}<\ln(1+\frac{1}{x})<\frac{1}{x}\ \ (x>0)\qquad\frac{x}{1+x}<\ln(1+x)<x\ \ (x>0)$$
$$\color{blue}{\text{注：}\Delta>0\text{前提下，灵活替换}x}$$

$$\text{柯西不等式：}$$
$$(ac+bd)^2\leqslant(a^2+b^2)(c^2+d^2)(\text{等号成立条件}\frac{a}{c}=\frac{b}{d};\quad\text{高维同样成立})$$

双阶乘

$$(2n)!!\quad(2n-1)!!$$

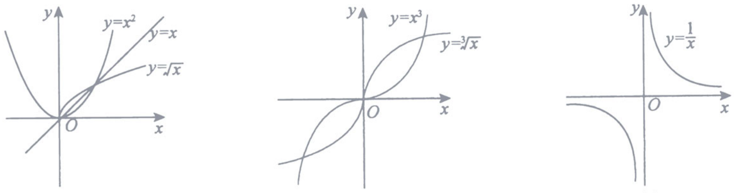
$$(2n)!!=2\cdot4\cdot6\cdots=2^n\cdot n!$$

$$(2n-1)!!=1\cdot3\cdot5\cdots$$

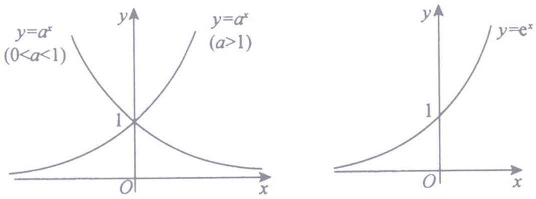
函数图像1

$$x^u(u=-1,\frac{1}{2},\frac{1}{3},1,2,3)$$

$$a^x(a>0\text{且}a\neq 1),e^x$$
$$\log_a x(a>0\text{且}a\neq 1),\ln x$$
$$\sin x,\cos x$$



(a) (b) (c)



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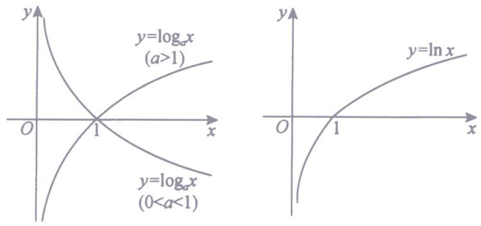

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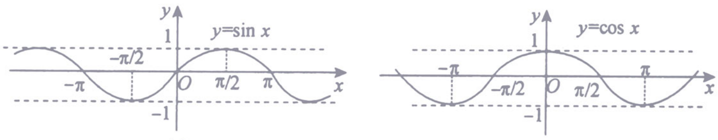
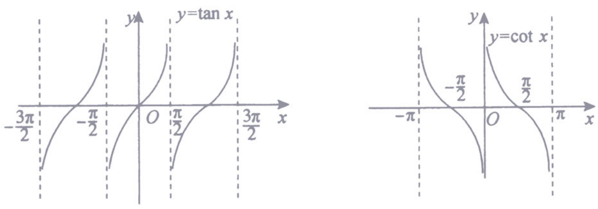


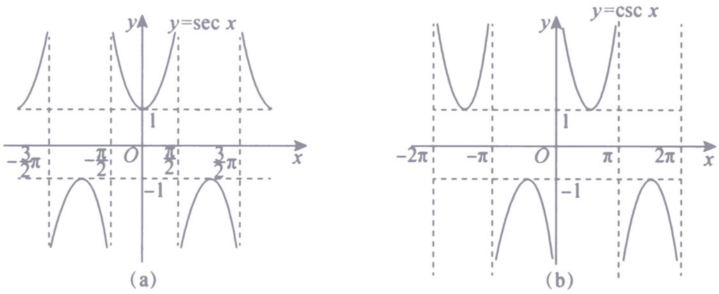
(a) (b)

函数图像2

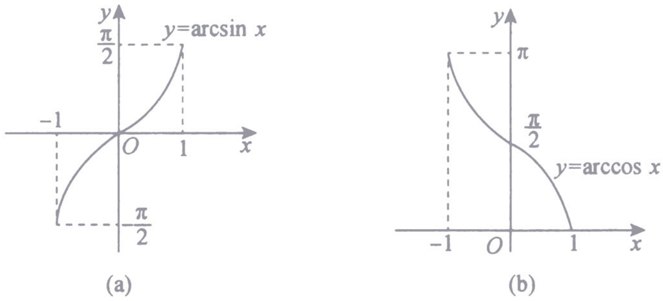
$$\tan x,\cot x$$
$$\sec x,\csc x$$
$$\arcsin x,\arccos x$$
$$\arctan x,arccotx$$

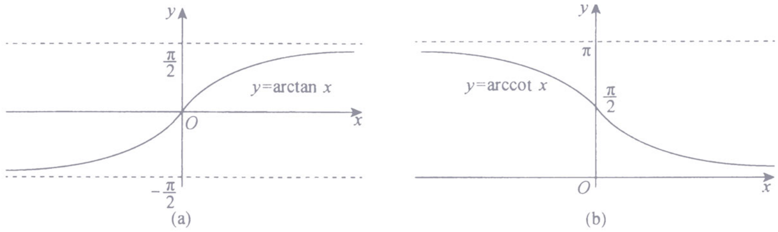


(a) (b)



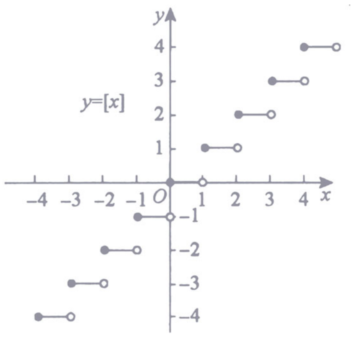
(a) (b)





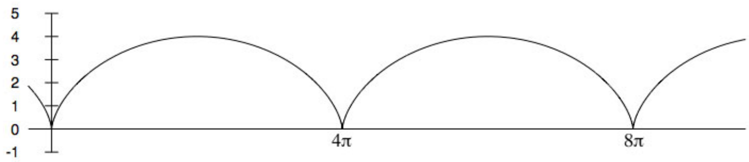
取整函数

$$x - 1 < [x] \leq x$$



摆线

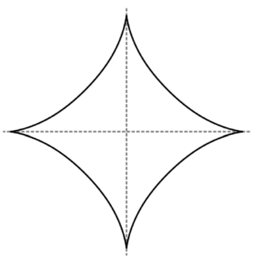
$$\begin{cases} x = 2(t - \sin t) \\ y = 2(1 - \cos t) \end{cases} \qquad a = 2\text{时}$$



a代表半径，t代表旋转角度，几何意义$\begin{cases} x = 2t - 2\sin t \\ y = 2 - 2\cos t \end{cases}$

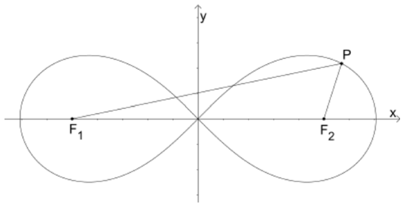
星形线

$$\begin{cases} x = a\cos^3 t \\ y = a\sin^3 t \end{cases} \qquad \text{顶点距离原点为}a$$



双纽线

直角坐标：$(x^2+y^2)^2=2a^2(x^2-y^2)$
极坐标：$r^2=2a^2\cos 2\theta$



常见几何计算公式

$$V_{球}=\frac{4}{3}\pi r^3 \quad S_{球}=4\pi r^2$$

$$V_{圆锥}=\frac{1}{3}\pi r^2 h \quad S_{圆锥侧}=\pi r l \quad V_{锥体}=\frac{1}{3}Sh\text{（一般圆锥，棱锥）}$$

$$V_{椭球}=\frac{4}{3}\pi abc \quad S_{椭圆}=\pi ab$$

$$S_{扇}=\frac{1}{2}\rho^2\theta$$

过曲线外一点做切线，求切线

曲线$y=\ln x$，设切点为$(x,\ln x)$

$Y-\ln x=\dfrac{1}{x}(X-x)$，代入点$(x_0,y_0)$，解$x$即可